Jan Gugenheimer

(curriculum vitae)

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Research Experience

since June 2022 **W2 Assistant Professor (TT)**, *TU Darmstadt*, Darmstadt.
Heading the Human-Computer Interaction Group at the TU Darmstadt

01/2020-05/2022 **Assistant Professor (MCF)**, *Télécom Paris/Institute Polytechnique de Paris*, Paris.
Teaching and researching at the intersection of Mixed Reality and Human-Computer Interaction

05/2017-07/2017 **Research Intern**, *Microsoft Research*, Redmond, Supervisor: Hvroje Benko.
I was working within the EPIC group at Microsoft Research in Redmond. My project focused on enabling true mobile virtual reality.

10/2015-1/2016 **Visiting Researcher**, *MIT*, Cambridge, Supervisor: Prof. Dr. Pattie Maes.
I was working at the Fluid Interfaces group at the MIT Media Lab. My research focused on new feedback concepts for mobile VR HMDs. The results were published at UIST'16 [Conf47]

10/2013-03/2014 **Visiting Researcher**, *ETH*, Zurich, Supervisor: Prof. Dr. Elgar Fleisch.
Working on ProductKing, a mobile application for consumer-driven product loyalty in the context of the Internet of Things.

04/2013-09/2013 **Master Thesis**, *Mercedes-Benz Research North America*, Palo Alto.
I worked together with the Mercedes-Benz HCI Research team on my master thesis. The focus was on an in-car authentication system using the rotary control knob. The results were published at AutomotiveUI '14 [Conf54]

Education

2014–2019 **Ph.D.**, *Ulm University*, Ulm, Supervisor: Prof. Dr. Enrico Rukzio.
Nomadic VR: Overcoming Challenges for Mobile Virtual Reality Head-Mounted Displays
Reviewer: *Prof. Dr. Albrecht Schmidt and Prof. Dr. Timo Ropinski*
Submitted: *1st April 2019*, Defense Date: *24th July 2019*

2011–2013 **Master of Science**, *Ulm University*, Ulm.
Studies of Media Informatics Grade: 1.0 / GPA: 4.0

2011–2012 **Exchange Student**, *Hawaii Pacific University*, Honolulu.
Information Systems Grade: 1.0 / GPA: 4.0

2007–2011 **Bachelor of Science**, *Ulm University*, Ulm.
Studies of Media Informatics Grade: 1.4 / GPA: 3.7

Funding & Awards

Personal Funding

2024

Reasonable AI, *Cluster of Excellence (DFG)*, **approx. 50,000,000 Euro (total cluster)**, Co-PI in the Systemic AI area.
Duration: 7 Years Student: Yanni Zhang

2024

XR-Guard, *BMBF*, **1,713,000 Euro**, I am the coordinator (lead PI) and PI of the project "XR-Guard: A Security Mechanism to Detect and Mitigate Perceptual Manipulations in Extended Reality" at the ATHENE Research Center for Cybersecurity.
Student: Jonas Wombacher

2024

EmergenCity, *LOEWE*, **342,123 Euro**, A collaborative project (Co-PI) inside of the state of Hessen, financing 30 PIs to work and research topics around resilient digital infrastructure.
Student: Yanni Mei

2024
2021

InterPlay, *ANR*, **165,809 Euro**, An ANR grant in collaboration (Co-PI) with eight researchers (my responsibility is one of three work packages financing one Phd student for 3 years) "InterPlay: Interacting with Rich Display Ecosystems".
Returned after leaving France

2024
2021

HARMFULVR, *ANR JCJC*, **250,000 Euro**, A young researcher ANR grant proposal (lead PI) funding one PhD student, one post-doc and multiple interns on the topic of "Assessing the Potential for Physiological Harm of Persuasive Interaction Design in Virtual Reality".
Returned after leaving France

2024
2021

IMT, *Futur & Ruptures*, **109,713 Euro**, A PhD Grant to fund a 3 Year PhD Student on the topic of "MEMOVR: Inducing and Protecting against False Memories through Interaction and Interface Design in Virtual Reality.".
Student: Elise Bonnail

2023
2020

IP-Paris, *Telecom Paris*, **109,713 Euro**, A PhD Grant to fund a 3 Year PhD Student on the topic of "Augmented Humanity: Exploring Ethical, Social and Political Challenges of Mixed Reality using Speculative Design and Design Fiction.".
Student: Wen-Jie Tseng, now funded at TU-Darmstadt

Paper Awards

Best Paper, *MobileHCI 2023[Conf19],CHI 2017[Conf45]*.

Best Paper Honorable Mention, *CHI 2025[Conf9], CHI 2023[Conf20] , CHI 2022[Conf26], CHI 2020[Conf35] , ITS 2014[Conf53]*.

Demo Award, *IEEE VR 2018, CHI 2025*, Best Demo Honorable Mention Award for ShareVR [Adj21] (IEEE VR 2018) and SwitchAR [Adj4] (CHI 2025).

Teaching

Teaching at TU-Darmstadt

since 2022 **Interaction in Virtual and Augmented Reality**, *TU-Darmstadt*, Lecture, An introduction to Augmented/Virtual and Mixed Reality challenges from an Human-Computer Interaction Perspective. **6 ECTS**.

since 2023 **Human-Computer Interaction**, *TU Darmstadt*, Lecture, An introduction class to Human-Computer Interaction **4 ECTS**.

since 2024 **Forschungsprojekt HCI**, *TU Darmstadt*, Project, A research project for students in HCI **8-10 ECTS**.

since 2026 **HCI Seminar**, *TU Darmstadt*, Seminar, Research trends in HCI and academic presentation (with Prof. Florian Müller).

Teaching at IP-Paris

2020-2022 **Human-Computer Interaction for Mixed Reality**, *Telecom-Paris*, Lecture, An introduction to Augmented/Virtual and Mixed Reality challenges from an Human-Computer Interaction Perspective. **2.5 ECTS**.

2020-2022 **Introduction to Human-Computer Interaction**, *TelecomSudParis*, Lecture, An introduction to basic concepts of Human-Computer Interaction. **3 ECTS**.

2020-2022 **Guided Research Project**, *Telecom-Paris*, Project, This module allows students to experience and be involved in ongoing research projects inside of teams of IP-Paris. **5-10 ECTS**.

2020-2022 **HCI/Graphics Project Seminar**, *Telecom-Paris*, Project, Student project in IGR 205 in the field of Human-Computer Interaction and Mixed Reality. **2.5 ECTS**.

Teaching at Ulm University

2014-2019 **Lectures**, *Ulm University*, I was giving lectures for classes such as: User Interface Softwaretechnology, Digital Media and Mobile Human-Computer Interaction.

2014-2019 **Seminar**, *Ulm University*, WS '14, WS '16, WS '19, Research Trends in Media Informatics.

2019 **Proseminar**, *Ulm University*, SS '19, Human-Computer Interaction.

2014-2015 **Project**, *Ulm University*, WS '14 - SS '15, Project Human-Computer Interaction.

Leadership

Leadership Seminars

2023-2025 **Seminar**, *ProFührung*, TU Darmstadt, Leadership development for professors.

Supervising PhD Students

2025-now **PhD Thesis**, *Yanni Zhang*, Immersive Human-Agent Interaction.

2025-now **PhD Thesis**, *Kian Mirbagheri*, Immersive Deceptive Interactions.

2024-now **PhD Thesis**, *Jonas Wombacher*, Safeguarding Perceptual Manipulations in Augmented Reality.

2023-now **PhD Thesis**, *Yanni Mei*, Immersive Visualization of Future Events in Extended Reality.

2023-now **PhD Thesis**, *Kevin Linne*, Interaction Across Multiple Realities.

2023-now **PhD Thesis**, *Martin Pluisch*, Context Adaptive Augmented Reality.

Graduated PhD Students/Postdocs

2024-2025 **Postdoctoral Researcher**, *Dr. Veronika Krauss*, Deceptive Designs in Extended Reality and AI. .
Now: Professor at Ansbach University

2022-2025 **PhD Thesis**, *Elise Bonnail*, Memory Distortions Induced by Reality-Altering Media. **[Co-Supervised with Dr. Samuel Huron and Prof. Dr. Eric Lecolinet]**.
Now: Lil Weasel Paris

2021-2025 **PhD Thesis**, *Wen-Jie Tseng*, Being in Two Spaces: Investigating and Mitigating Spatial Conflicts in Virtual Reality. **[Co-Supervised with Dr. Samuel Huron and Prof. Dr. Eric Lecolinet]**.
Now: Postdoc at Sorbonne Université

2019-2024 **PhD Thesis**, *Jan Ole Rixen*, Social Augmented Reality: Human-Based Augmentations and their Effects on Augmenter and Augmented. **[Co-Supervised at Ulm University with Prof. Dr. Enrico Rukzio]**.
Now: Postdoc at Karlsruhe Institute of Technology

2018-2024 **PhD Thesis**, *Evgeny Stemasov*, Ubiquitous Personal Fabrication: Modeling-Free Instead of Increasingly Simple. **[Co-Supervised at Ulm University with Prof. Dr. Enrico Rukzio]**.
Now: Senior Research Scientist at Autodesk Research Toronto

Recent Academic Service & Voluntary Activities

Organizational Member

German PreCHI 2027 **Chair**, Darmstadt.

CHI 2025 **Organizer**, *Workshop: Developing Sociotechnical Solutions to Mitigate New Harms in Immersive and Embodied Virtual Spaces*, Yokohama.

CSCW 2024 **Organizer**, *Workshop: Understanding and Mitigating New Harms in Immersive and Embodied Virtual Spaces*, San José, Costa Rica.

CHI 2024 **Organizer**, *Workshop: Novel Approaches for Understanding and Mitigating Emerging New Harms in Immersive and Embodied Virtual Spaces*, Honolulu.

ISS 2023 **Doctoral Symposium Co-Chair**, *ACM Interactive Surfaces and Spaces.*

Mensch und Computer 2024 **Demo Co-Chair**, Karlsruhe.

Mensch und Computer 2022 — **Student Research Competition Co-Chair**, Darmstadt.

SSXPR 2022 — **Organizer**, *Workshop on Novel Challenges of Safety, Security and Privacy in Extended Reality at CHI 2022*, New Orleans.

Cross-Reality Interaction 2020 — **Organizer**, *Workshop at ISS 2020*, Lisbon.

PerDis 2020 — **Program Co-Chair**, *International Symposium on Pervasive Displays*, Manchester.

EthicalHMDs 2020 — **Organizer**, *Workshop at CHI 2020*, Honolulu.

SocialHMDs 2019 — **Organizer**, *Workshop at CHI 2019*, Glasgow.

UIST 2018 — **Web Chair**, *ACM User Interface Software and Technology Symposium*, Berlin.

ISCT 2015 — **Local Chair**, *International Symposium on Companion Technology*, Ulm.

Program Commitee Member

since 2021 — **UIST**, *ACM User Interface Software and Technology Symposium*.

'21 — **IEEE VR**, *IEEE Conference on Virtual Reality and 3D User Interfaces*.

since 2020 — **CHI**, *ACM Conference on Human Factors in Computing Systems*.

'20,'21 — **AHs**, *Augmented Humans International Conference 2021*.

'19 — **ISS**, *ACM Conference on Interactive Surfaces and Spaces*.

'17,'18,'19 — **LBW CHI**, *ACM Conference on Human Factors in Computing Systems*.

'18 — **PerDis**, *ACM International Symposium on Pervasive Displays*.

'17,'19,'20 — **MUM**, *International Conference on Mobile and Ubiquitous Multimedia*.

'19,'20 — **MuC**, *Mensch und Computer*.

'19,'20 — **VARECo**, *Workshop on VR and AR in Everyday Context at Mensch und Computer*.

Reviewer

since 2015 — **CHI**, *ACM Conference on Human Factors in Computing Systems*.

since 2017 — **UIST**, *ACM User Interface Software and Technology Symposium*.

since 2020 — **IMWUT**, *Interactive, Mobile, Wearable and Ubiquitous Technologies*.

since 2018 — **ToCHI**, *ACM Transactions on Computer-Human Interaction*.

'17,'18 — **IEEE VR**, *IEEE Conference on Virtual Reality and 3D User Interfaces*.

'17,'20 — **SIGGRAPH**, *ACM Special Interest Group on Computer Graphics*.

'17 — **DIS**, *ACM International Conference on Designing Interactive Systems*.

'16, '19 — **CHI Play**, *ACM Annual Symposium on Computer-Human Interaction in Play*.

'15 — **MobileHCI**, *ACM International Conference on Human-Computer Interaction with Mobile Devices and Services*.

'14,'15 — **ITS/ISS**, *ACM International Conference on Interactive Surfaces and Spaces*.

'18 — **NordiCHI**, *Nordic forum for Human-Computer Interaction*.

'16 — **TEI**, *ACM International Conference on Tangible, Embedded, and Embodied Interactions*.

'15,'16 — **AH**, *ACM Augmented Human International Conference*.

'14,'15 — **AutoUI**, *ACM Conference on Automotive User Interfaces and Interactive Vehicular Applications*.

Selected Invitations

2026 **NII Shonan Meeting**, *Invited participant*, Japan.

2026 **Aarhus XR Summer School**, *Lecturer*, Aarhus University.

2024 **XR Summer School**, *Lecturer*, CWI Amsterdam.

2023 **Dagstuhl Seminar on XR**, *Invited participant*, Schloss Dagstuhl.

2023 **XR Summer School**, *Lecturer*, University of Copenhagen.

2022 **Digital Tech Summit**, *Invited speaker*, Copenhagen.

Selected Media Coverage

2025 **idw – Informationsdienst Wissenschaft**, *Studie zeigt: KI erzeugt manipulative Designmuster in Webseiten*.

2025 **PAGE online**, *Dark Patterns und KI in der Website-Entwicklung*.

2025 **https://theministryofai.org/**, *Unmasking the AI Dilemma: When ChatGPT's Web Designs Turn Deceptive Without Anyone Noticing*.

2024 **Eva Wolfangel**, *Manipuliertes Gedächtnis English: Manipulated Memory*.

2023 **New Scientist**, *Play VR games on a bus by wiggling your fingers as if they were arms*.

2022 **FIF Podcast**, *Herausforderungen des Metaversums, English: Challenges of the Metaverse*.

2022 **Le Monde**, *La réalité virtuelle face au risque de détournement malveillant de ses outils, English: Virtual reality faces the risk of malicious misuse of its tools "*.

2020 **Telecom-Paris**, *Spatial Computing, the next paradigm of Human-Computer Interaction*.

2020 **I'MTech Blog**, *Quelles seront nos interactions avec la réalité virtuelle ?*.

All Publications

Conference Proceedings (Full and Short Paper)

[Conf1] Yanni Mei, Samuel Wendt, Jonas Wombacher, Florian Müller, and **Gugenheimer, Jan**. Shadar: Llm-driven shader generation to transform visual perception in augmented reality. In *Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST '26)*. ACM, 2026. Accepted, to appear.

[Conf2] Martin Pluisch, **Gugenheimer, Jan**, Youngjun Cho, Simon Julier, and Ernst Kruijff. A multi-sensor approach for cognitive load assessment in mobile augmented reality. In *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 208–218. IEEE, 2025.

[Conf3] Yanni Mei, Samuel Wendt, Florian Müller, and **Gugenheimer, Jan**. Meme, myself and ar: Exploring memes sharing in face-to-face conversation using augmented reality. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI '26, New York, NY, USA, 2026. Association for Computing Machinery.

[Conf4] Maarten Gerritse, Iris Folpmers, Caesar Alpha Irawan, Guo Freeman, **Gugenheimer, Jan**, and Julian Frommel. Classifying harms in social vr: A systematic literature review and taxonomy. *Proc. ACM Hum.-Comput. Interact.*, 10(1), March 2026.

[Conf5] Yanni Mei, Jonas Wombacher, Wen-Jie Tseng, Veronika Krauß, and **Gugenheimer, Jan**. Embodied spatial simulation: Enabling embodied interaction with human motion simulation in extended reality. *IEEE Transactions on Visualization and Computer Graphics*, 32(5):4154–4164, 2026.

[Conf6] Martin Pluisch, **Gugenheimer, Jan**, and Ernst Kruijff. Context-aware application recommendations in augmented reality. In *Proceedings of the 2025 ACM Symposium on Spatial User Interaction*, SUI '25, New York, NY, USA, 2025. Association for Computing Machinery.

[Conf7] Jonas Wombacher, Zhipeng Li, and **Gugenheimer, Jan**. Switchar: Perceptual manipulations in augmented reality. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, UIST '25, New York, NY, USA, 2025. Association for Computing Machinery.

[Conf8] Julius von Willich, Frank Nelles, Wen-Jie Tseng, **Gugenheimer, Jan**, Sebastian Günther, and Max Mühlhäuser. A qualitative investigation of user transitions and frictions in cross-reality applications. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, New York, NY, USA, 2025. Association for Computing Machinery.

[Conf9] Veronika Krauß, Mark McGill, Thomas Kosch, Yolanda Maira Thiel, Dominik Schön, and **Gugenheimer, Jan**. "create a fear of missing out" - chatgpt implements unsolicited deceptive designs in generated websites without warning. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, New York, NY, USA, 2025. Association for Computing Machinery.

[Conf10] Xiang Li, Wei He, Shan Jin, **Gugenheimer, Jan**, Pan Hui, Hai-Ning Liang, and Per Ola Kristensson. Investigating creation perspectives and icon placement preferences for on-body menus in virtual reality. *Proc. ACM Hum.-Comput. Interact.*, 8(ISS), October 2024.

[Conf11] Martina Ruocco, Pejman Saeghe, Frederic Kerber, **Gugenheimer, Jan**, Mark McGill, and Mohamed Khamis. From redirected navigation to forced attention: Uncovering manipulative and deceptive designs in augmented reality through retail shopping. In *Proceedings of the International Symposium on Mixed and Augmented Reality*, 2024.

[Conf12] Elise Bonnail, Julian Frommel, Eric Lecolinet, Samuel Huron, and **Gugenheimer, Jan**. Was it real or virtual? confirming the occurrence and explaining causes of memory source confusion between reality and virtual reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–17, 2024.

[Conf13] Evgeny Stemasov, Simon Demharter, Max Rädler, **Gugenheimer, Jan**, and Enrico Rukzio. param: Leveraging parametric design in extended reality to support the personalization of artifacts for personal fabrication. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–22, 2024.

[Conf14] Evgeny Stemasov, Tobias Wagner, Ali Askari, Jessica Janek, Omid Rajabi, Anja Schikorr, Julian Frommel, **Gugenheimer, Jan**, and Enrico Rukzio. Dungeonmaker: Embedding tangible creation and destruction in hybrid board games through personal fabrication technology. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–20, 2024.

[Conf15] Wen-Jie Tseng, Petros Dimitrios Kontrazis, Eric Lecolinet, Samuel Huron, and **Gugenheimer, Jan**. Understanding interaction and breakouts of safety boundaries in virtual reality through mixed-method studies. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 482–492. IEEE, 2024.

[Conf16] Jan Ole Rixen, Jan Henry Belz, Luca-Maxim Meinhardt, **Gugenheimer, Jan**, and Enrico Rukzio. Exploring the effects of head-mounted augmented reality on helping behaviour. In

Proceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia, pages 440–455, 2023.

[Conf17] Joseph O'Hagan, **Gugenheimer, Jan**, Jolie Bonner, Florian Mathis, and Mark McGill. Augmenting people, places & media: The societal harms posed by everyday augmented reality, and the case for perceptual human rights. In *Proceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia*, pages 225–235, 2023.

[Conf18] Evgeny Stemasov, Jessica Hohn, Maurice Cordts, Anja Schikorr, Enrico Rukzio, and **Gugenheimer, Jan**. Brickstart: Enabling in-situ design and tangible exploration for personal fabrication using mixed reality. *Proceedings of the ACM on Human-Computer Interaction*, 7(ISS):64–92, 2023.

[Conf19] Jan Ole Rixen, Luca-Maxim Meinhardt, Michael Glöckler, Marius-Lukas Ziegenbein, Anna Schlothauer, Mark Colley, Enrico Rukzio, and **Gugenheimer, Jan**. The loop and reasons to break it: Investigating infinite scrolling behaviour in social media applications and reasons to stop. *Proceedings of the ACM on Human-Computer Interaction*, 7(MHCI):1–22, 2023.

[Conf20] Elise Bonnail, Wen-Jie Tseng, Mark McGill, Eric Lecolinet, Samuel Huron, and **Gugenheimer, Jan**. Memory manipulations in extended reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–20, 2023.

[Conf21] Wen-Jie Tseng, Samuel Huron, Eric Lecolinet, and **Gugenheimer, Jan**. Fingermapper: Mapping finger motions onto virtual arms to enable safe virtual reality interaction in confined spaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–14, 2023.

[Conf22] Yi Fei Cheng, Hang Yin, Yukang Yan, **Gugenheimer, Jan**, and David Lindlbauer. Towards understanding diminished reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems **CHI 2022***, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf23] Evgeny Stemasov, Tobias Wagner, **Gugenheimer, Jan**, and Enrico Rukzio. Shapefindar: Exploring in-situ spatial search for physical artifact retrieval using mixed reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems **CHI 2022***, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf24] Jan Ole Rixen, Mark Colley, Ali Askari, **Gugenheimer, Jan**, and Enrico Rukzio. Consent in the age of ar: Investigating the comfort with displaying personal information in augmented reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems **CHI 2022***, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf25] Wen-Jie Tseng, Elise Bonnail, Mark McGill, Mohamed Khamis, Eric Lecolinet, Samuel Huron, and **Gugenheimer, Jan**. The dark side of perceptual manipulations in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems **CHI 2022***, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf26] Uwe Gruenefeld, Jonas Auda, Florian Mathis, Stefan Schneegass, Mohamed Khamis, **Gugenheimer, Jan**, and Sven Mayer. Vrception: Rapid prototyping of cross-reality systems in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems **CHI 2022, *Best Paper Honorable Mention****, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf27] Evgeny Stemasov, Alexander Botner, Enrico Rukzio, and **Gugenheimer, Jan**. Ephemeral fabrication: Exploring a ubiquitous fabrication scenario of low-effort, in-situ creation of short-lived physical artifacts. In *Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction **TEI 2022***, TEI '22, New York, NY, USA, 2022. Association for Computing Machinery.

[Conf28] Jan Ole Rixen, Teresa Hirzle, Mark Colley, Yannick Etzel, Enrico Rukzio, and **Gugenheimer, Jan**. Exploring augmented visual alterations in interpersonal communication. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems **CHI 2021***.

[Conf29] Leon Müller, Ken Pfeuffer, **Gugenheimer, Jan**, Sarah Prange, Florian Alt, and Bastian Pfleging. Spatialproto: Using real-world captures for rapid prototyping of mixed reality experiences. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***CHI 2021***.

[Conf30] Teresa Hirzle, Maurice Cordts, Enrico Rukzio, **Gugenheimer, Jan**, and Andreas Bulling. A critical assessment of the use of ssq as a measure of general discomfort in vr head-mounted displays. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***CHI 2021***.

[Conf31] Pascal Jansen, Fabian Fischbach, **Gugenheimer, Jan**, Evgeny Stemasov, Julian Frommel, and Enrico Rukzio. Share: Enabling co-located asymmetric multi-user interaction for augmented reality head-mounted displays. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* ***UIST 2020***, page 459–471, New York, NY, USA, 2020. Association for Computing Machinery.

[Conf32] Gabriel Haas, **Gugenheimer, Jan**, and Enrico Rukzio. Voicemessage++: Augmented voice recordings for mobile instant messaging. In *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services* ***MobileHCI 2020***, New York, NY, USA, 2020. Association for Computing Machinery.

[Conf33] Gabriel Haas, **Gugenheimer, Jan**, Jan Ole Rixen, Florian Schaub, and Enrico Rukzio. "they like to hear my voice": Exploring usage behavior in speech-based mobile instant messaging. In *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services* ***MobileHCI 2020***, MobileHCI '20, New York, NY, USA, 2020. Association for Computing Machinery.

[Conf34] Mark Colley, Stefanos Can Mytilineos, Marcel Walch, **Gugenheimer, Jan**, and Enrico Rukzio. Evaluating highly automated trucks as signaling lights. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* ***AutomotiveUI 2020***, page 111–121, New York, NY, USA, 2020. Association for Computing Machinery.

[Conf35] Mark Colley, Marcel Walch, **Gugenheimer, Jan**, Ali Askari, and Enrico Rukzio. Towards inclusive external communication of autonomous vehicles for pedestrians with vision impairments. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***(CHI 2020, 10 Pages *Best Paper Honorable Mention*)***. ACM, 2020.

[Conf36] Evgeny Stemasov, Tobias Wagner, **Gugenheimer, Jan**, and Enrico Rukzio. Mix&match: Towards omitting modelling through in-situ alteration and remixing of model repository artifacts in mixed reality. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***(CHI 2020, 8.5 Pages)***. ACM, 2020.

[Conf37] Dennis Wolf, **Gugenheimer, Jan**, Marco Combosch, and Enrico Rukzio. Understanding the heisenberg effect of spatial interaction: A selection induced error for spatially tracked input devices. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***(CHI 2020, 8.5 Pages)***. ACM, 2020.

[Conf38] Tobias Drey, **Gugenheimer, Jan**, Julian Karlbauer, Maximilian Milo, and Enrico Rukzio. Vrsketchin: Exploring the design space of pen and tablet interaction for 3d sketching in virtual reality. In *Proceedings of the 2020 Conference on Human Factors in Computing Systems* ***(CHI 2020, 10 Pages)***. ACM, 2020.

[Conf39] Teresa Hirzle, **Gugenheimer, Jan**, Florian Geiselhart, Andreas Bulling, and Enrico Rukzio. A design space for gaze interaction on head-mounted displays. In *Proceedings of the 2019 Conference on Human Factors in Computing Systems* ***(CHI 2019, 12 Pages)***. ACM, 2019.

[Conf40] Michael Rietzler, **Gugenheimer, Jan**, Teresa Hirzle, Martin Deubzer, Eike Langbehn, and Enrico Rukzio. Rethinking redirected walking: On the use of curvature gains beyond perceptual limitations and revisiting bending gains. In *Proceedings of the IEEE International Symposium for Mixed and Augmented Reality* ***(ISMAR 2018, 8 Pages)***. ACM, 2018.

[Conf41] Michael Rietzler, Teresa Hirzle, **Gugenheimer, Jan**, Julian Frommel, Thomas Dreja, and Enrico Rukzio. Vrspinning: Exploring the design space of a 1d rotation platform to increase the perception of self-motion in vr. In *Proceedings of the Conference on Designing Interactive Systems* ***(DIS 2018, 10 Pages)***. ACM, 2018.

[Conf42] Michael Rietzler, Florian Geiselhart, **Gugenheimer, Jan**, and Enrico Rukzio. Breaking the tracking: Enabling weight perception using perceivable tracking offsets. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* ***(CHI 2018, 12 Pages)***, page 128. ACM, 2018.

[Conf43] **Gugenheimer, Jan**, Evgeny Stemasov, Harpreet Sareen, and Enrico Rukzio. Facedisplay: Towards asymmetric multi-user interaction for nomadic virtual reality. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* ***(CHI 2018, 13 Pages)***, page 54. ACM, 2018.

[Conf44] Philipp Hock, Sebastian Benedikter, **Gugenheimer, Jan**, and Enrico Rukzio. Carvr: Enabling in-car virtual reality entertainment. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* ***(CHI 2017, 11 Pages)***, pages 4034–4044. ACM, 2017.

[Conf45] **Gugenheimer, Jan**, Evgeny Stemasov, Julian Frommel, and Enrico Rukzio. Sharevr: Enabling co-located experiences for virtual reality between hmd and non-hmd users. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* ***(CHI 2017, 13 Pages *Best Paper Award*)***, pages 4021–4033. ACM, 2017.

[Conf46] **Gugenheimer, Jan**, Katrin Plaumann, Florian Schaub, Patrizia Di Campli San Vito, Saskia Duck, Melanie Rabus, and Enrico Rukzio. The impact of assistive technology on communication quality between deaf and hearing individuals. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* ***(CSCW 2017, 14 Pages)***, pages 669–682. ACM, 2017.

[Conf47] **Gugenheimer, Jan**, Dennis Wolf, Eythor R Eiriksson, Pattie Maes, and Enrico Rukzio. Gyrovr: Simulating inertia in virtual reality using head worn flywheels. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* ***(UIST 2016, 6 Pages)***, pages 227–232. ACM, 2016.

[Conf48] **Gugenheimer, Jan**, David Dobbelstein, Christian Winkler, Gabriel Haas, and Enrico Rukzio. Facetouch: Enabling touch interaction in display fixed uis for mobile virtual reality. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* ***(UIST 2016, 12 Pages)***, pages 49–60. ACM, 2016.

[Conf49] **Gugenheimer, Jan**, Dennis Wolf, Gabriel Haas, Sebastian Krebs, and Enrico Rukzio. Swivrchair: A motorized swivel chair to nudge users' orientation for 360 degree storytelling in virtual reality. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* ***(CHI 2016, 5 Pages)***, pages 1996–2000. ACM, 2016.

[Conf50] **Gugenheimer, Jan**, Alexander De Luca, Hayato Hess, Stefan Karg, Dennis Wolf, and Enrico Rukzio. Colorsnakes: Using colored decoys to secure authentication in sensitive contexts. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services* ***(MobileHCI 2015, 10 Pages)***, pages 274–283. ACM, 2015.

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